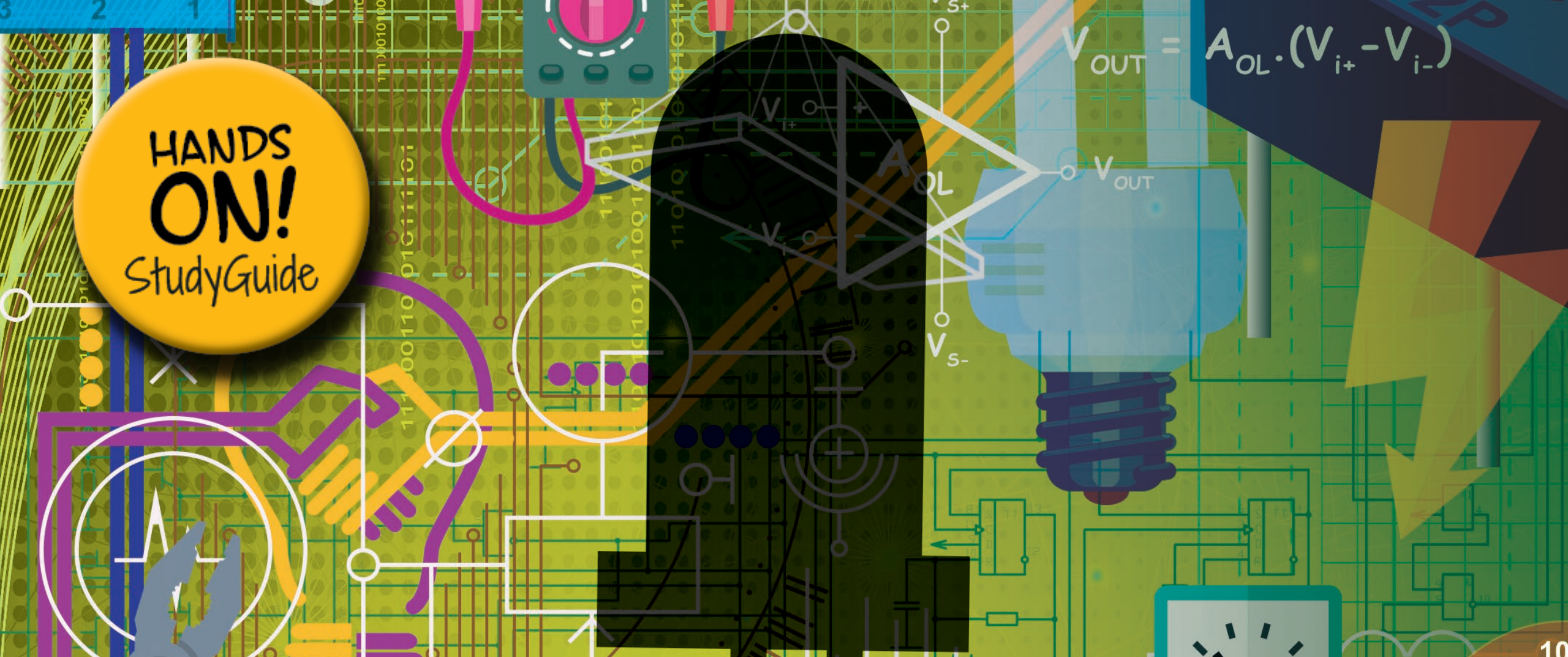




**HANDS
ON!**
StudyGuide

$$V_{OUT} = A_{OL} \cdot (V_{i+} - V_{i-})$$

Industrial Electronics

N5

Module 1: Transistor amplifiers

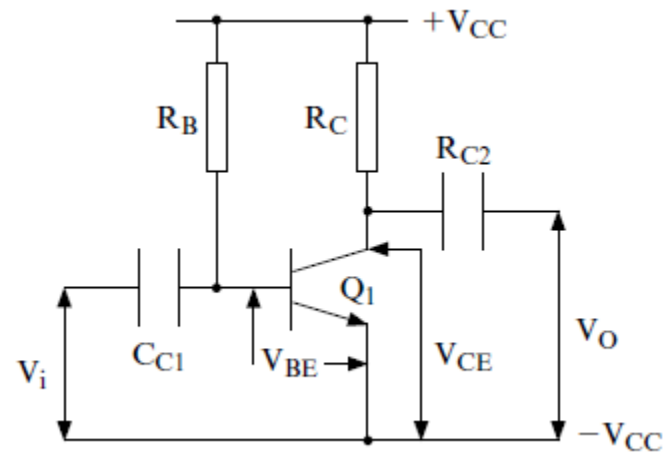
INTRODUCTION

Any transistor has a relation to current, voltage and impedance and is very often referred to as the parameters of the transistor. It is of utmost importance that circuits are 'matched' to one another in order to obtain maximum power transfer. This matching of circuits specifically refers to the impedances to be matched.

Module 1: Transistor amplifiers (continued)

FIXED BIAS COMMON EMITTER AMPLIFIER

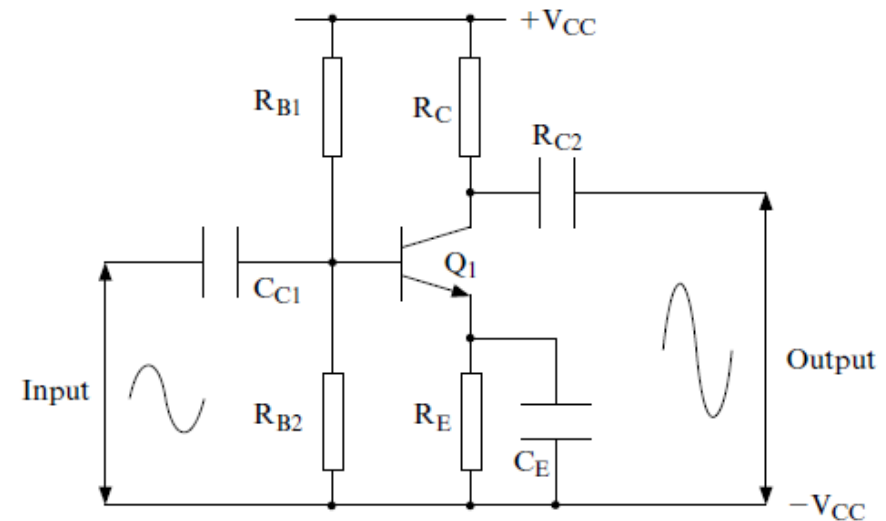
The common emitter amplifier circuit diagram illustrated in the figure below is that of a fixed bias configuration. This means that only one supply is used bias both the base and the collector.



Module 1: Transistor amplifiers (continued)

VOLTAGE DIVIDER BIAS COMMON EMITTER AMPLIFIER

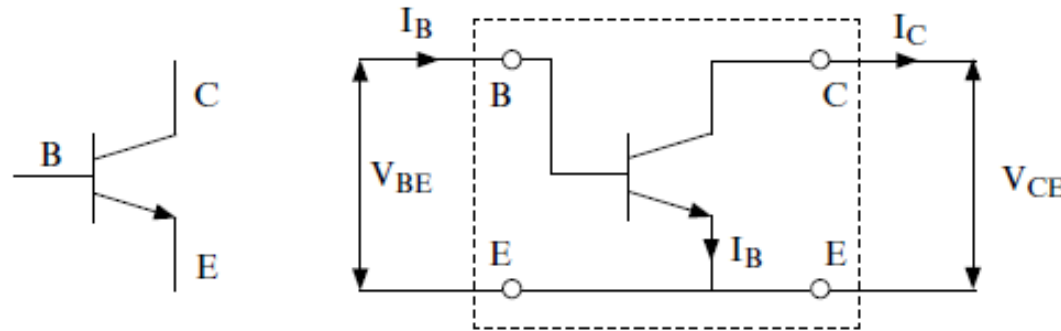
The common emitter amplifier circuit diagram illustrated in the figure below is that of a voltage divider bias configuration



Module 1: Transistor amplifiers (continued)

THE TRANSISTOR AS A TWO-PORT DEVICE

The transistor in the figure below illustrates a transistor as a two-port device in the common emitter configuration.



Module 1: Transistor amplifiers (continued)

SMALL SIGNAL ANALYSIS (COMMON EMITTER AND COMMON BIAS)

All amplifiers are two-port devices that consist of two input terminals and two output terminals. Five important characteristics to be calculated are:

- Input impedance;
- Output impedance;
- Current gain;
- Voltage gain; and
- Power gain.

Module 1: Transistor amplifiers (continued)

DISTORTION

Distortion is one of the major challenges when designing amplifiers.

Distortion can be described as not obtaining an exact amplified output signal version of that which we applied as input.

Module 2: Oscillators

INTRODUCTION

Oscillators form an integral part of the electronic industry since various types of waveforms for different applications can be generated by such circuits.

Oscillators are found as low frequency oscillators as well as high frequency oscillators. High frequency oscillators are extensively used in the radio and television communication field. Low frequency oscillators are mainly used in the electronic field, such as digital circuits, and for the testing of electronic circuits and test equipment.

Module 2: Oscillators (continued)

CRITERIA FOR AN OSCILLATOR

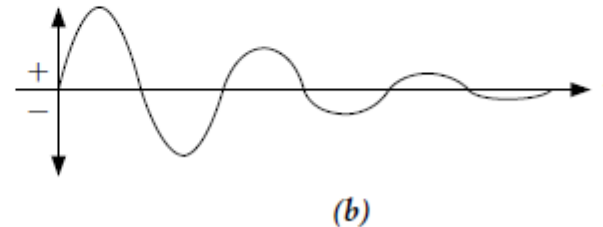
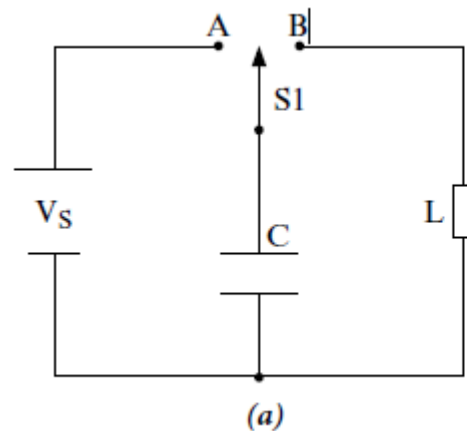
All oscillators, irrespective of the type of waveform produced, must at least conform to the following criteria:

- The frequency the oscillator produces must be very stable.
- The amplitude of the output must be constant.
- Provision must be made for positive feedback.

Module 2: Oscillators (continued)

THE TUNED CIRCUIT

An oscillator circuit will depend upon its operation on the basic principle of an LC-tuned circuit.



Module 2: Oscillators (continued)

THE BARKHAUSEN CRITERIA

$$A' = \frac{A}{1 + A \times \beta}$$

Where A' = overall gain with feedback

A = overall gain without feedback

β = feedback fraction

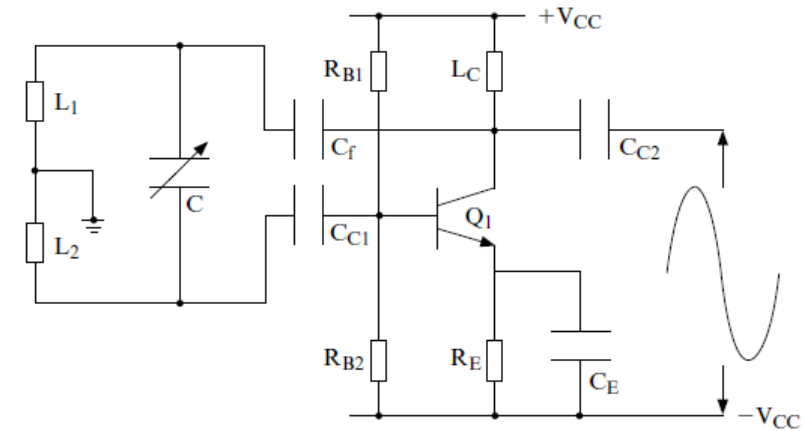
The Barkhausen criteria stipulates that the factor $A' \geq 1$ for self-sustained oscillations. Since we are dealing with positive feedback, it will be obvious that it will assist the output and our gain will actually increase

Module 2: Oscillators (continued)

OSCILLATOR CIRCUITS

The tuned circuit of a Hartley-oscillator consists of two inductors L_1 and L_2 and a tuning capacitor C_1 which will determine its frequency of operation.

The total inductance is given by $L_1 + L_2 + 2M$, where M is equal to the total conductance between the two inductors.



Module 2: Oscillators (continued)

MULTIVIBRATORS

Multivibrators are termed relaxation oscillators but with the difference that the output will be a square wave which comes in very useful in digital circuit applications. They come in different types:

- The A-stable multivibrator;
- The mono-stable multivibrator; and
- The bi-stable multivibrator.

Module 2: Oscillators (continued)

THE SCHMITT-TRIGGER

The Schmitt-trigger is mostly used in a frequency counter as a wave-shaper in that it is able to convert any type of waveform into a square wave and has the following characteristics:

- It is used as a wave-shaper and has two trigger points.
- It is a voltage-sensitive switch.
- The input and output frequencies are the same.
- It provides for a digital output.

Module 2: Oscillators (continued)

THE 555-PRECISION TIMER

A 555-precision timer is an integrated circuit utilising external components, such as resistors and capacitors, and is used as a timer that will produce square wave pulses for use where required.

Module 2: Oscillators (continued)

THE APPLICATION OF A 555-PRECISION TIMER

A 555-precision timer can be used to function for different operations:

Mono-stable application: Two external components need to be added to the 555-precision timer for it to function.

A-stable application: Three external components need to be added to the 555-precision timer for it to function.

A dc-dc voltage converter: Which is basically an A-stable multivibrator which has been modified or altered.

Module 3: Operational amplifiers

INTRODUCTION

All circuits have to be constructed using discrete components and that this is quite a cumbersome effort. The operational amplifier solves this problem to quite an extent in that an amplifier is now available in a single integrated circuit (IC) package.

Module 3: Operational amplifiers (continued)

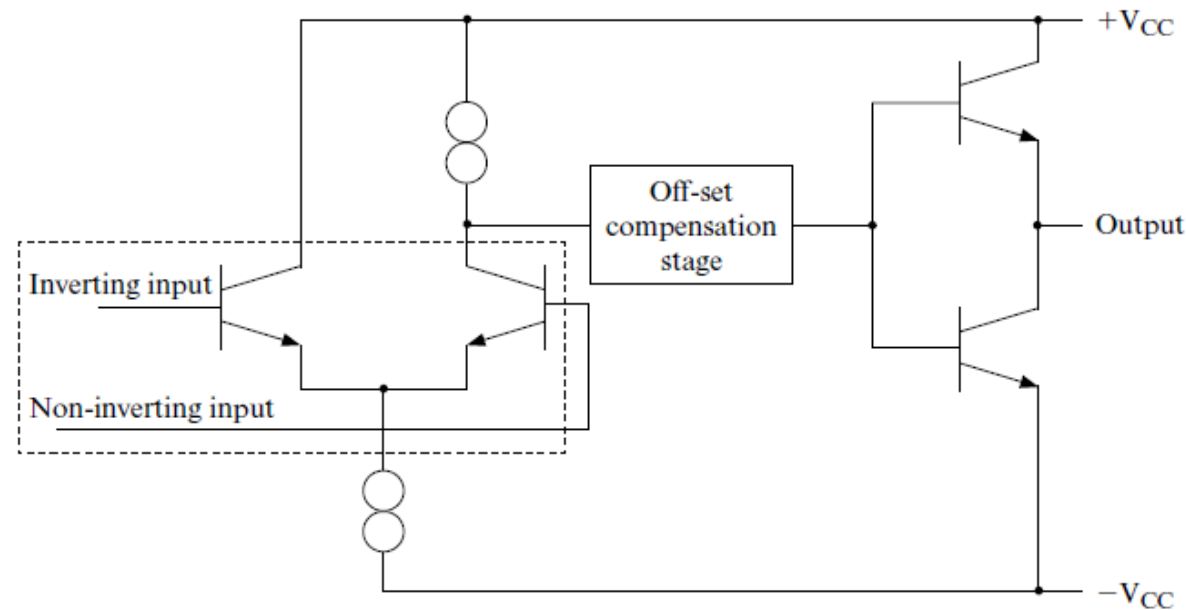
THE OPERATIONAL AMPLIFIER

Various modes of operation can be obtained from an operational amplifier and the operation is based on the concept of a differential amplifier. These modes of operation include amplifiers, active filters, voltage regulators and signal generators.

Module 3: Operational amplifiers (continued)

THE DIFFERENTIAL AMPLIFIER

A basic circuit of a differential amplifier is illustrated in the figure below:



Module 3: Operational amplifiers (continued)

MODES OF OPERATION

The good characteristics, size and cost of operational amplifiers makes them very versatile in a number of applications, such as:

The inverting amplifier;

The non-inverting amplifier;

The inverting summing amplifier (adder);

The voltage follower; and

The audio amplifier.

Module 3: Operational amplifiers (continued)

MATHEMATICAL FUNCTIONS USING OPERATIONAL AMPLIFIERS

The differentiator: $V_o(t) = \frac{-R_f C dV_i(t)}{dt}$

The integrator: $V_o(t) = \frac{-1(t)}{RC} \int V_i(t)$

Module 3: Operational amplifiers (continued)

THE COMPARATOR

A comparator is an electronic circuit utilising an operational amplifier that will compare a voltage supplied to it with a predetermined given reference voltage. The output that will be obtained will give an indication whether it is above or below the predetermined given reference voltage.

Module 3: Operational amplifiers (continued)

THE DIFFERENTIAL AMPLIFIER OR SUBTRACTOR

The differential amplifier differs from the comparator in the sense that the comparator will indicate the difference between a predetermined voltage level and a voltage that is applied to it, and the difference between two voltage levels of which the quantities are unknown. This type of circuit is mainly used where one will be required to monitor the level between two unknown or known voltage levels.

Module 3: Operational amplifiers (continued)

ACTIVE FILTERS

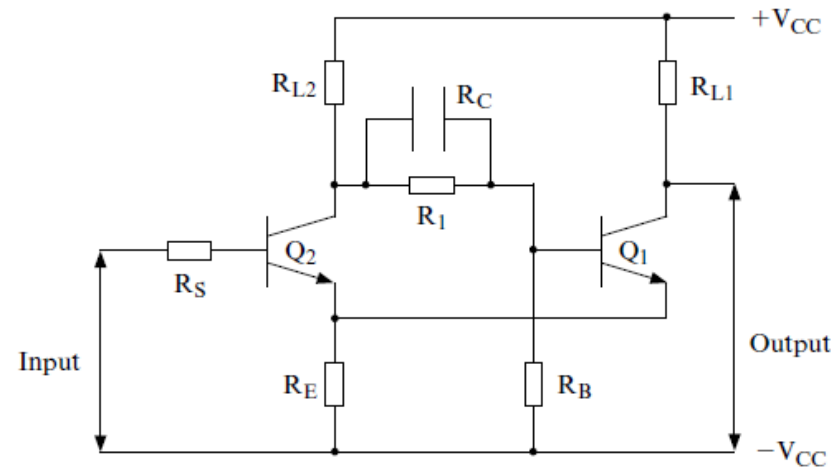
A filter can be defined as a device that will strongly discriminate against a specific frequency or a specific band of frequencies. There are four main types of filters that can be obtained:

- The low-pass filter;
- The high-pass filter;
- The band-pass filter; and
- The band-reject filter.

Module 4: Measuring instruments and test equipment

INSTRUMENT CIRCUITRY

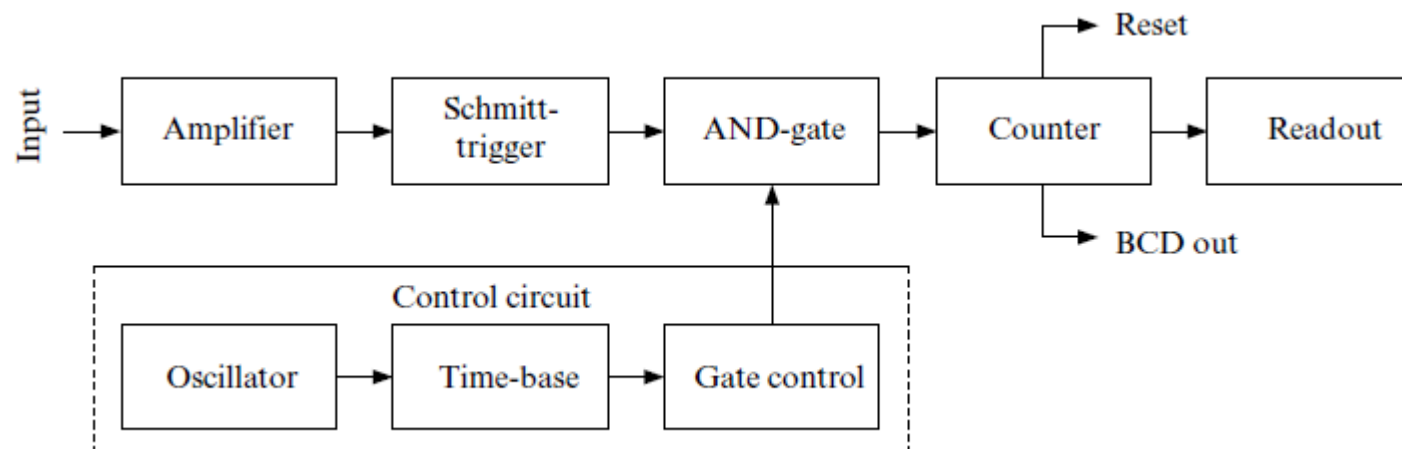
A Schmitt-trigger is a circuit used in a frequency counter to convert an analogue signal to a digital signal and is illustrated below.



Module 4: Measuring instruments and test equipment (continued)

A FREQUENCY COUNTER

A frequency counter is a device that will be able to measure frequency and is illustrated below:



Module 4: Measuring instruments and test equipment (continued)

DIGITAL VOLTMETERS

A digital voltmeter is a device that is able to measure the magnitude of an unknown voltage.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits

THE TIME CONSTANT IN AN RC-CIRCUIT

It is a known fact that a capacitor is a component that has the ability to store an electrical charge when subjected to a source of emf. There are three main factors that influence the capacity of a capacitor:

- Dielectric constant;
- Cross-sectional area of plates; and
- Distance between the plates.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

DIFFERENTIATORS AND INTEGRATORS

An RC-differentiator circuit, also termed an approximate differentiator, will have an output for a rectangular input which will be equal to the rate of change of the input voltage. An RC-integrator is obtained when the output is taken across the capacitor and what is most important is that the time constant be much greater than the mark period of the applied wave.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

THE TIME CONSTANT IN AN RL-CIRCUIT

An inductor is so manufactured that it will produce a specific amount of inductance and is dependent upon the following factors:

- The number of turns in the inductor;
- The cross-sectional area of the core;
- The permeability and ability to conduct magnetic flux of the core material;
and
- The length of the inductor.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

CHARGING AND DISCHARGING AN INDUCTOR

Just as much as an inductor is able to accumulate an electrical charge, so will it be able to lose an electrical charge. In electronic terms, we talk about an inductor discharging. It is known to us that it takes an inductor five time constants to build up a charge and that it only changes by 63,2% per time constant and the voltage will be decreasing and the charge current will be increasing.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

HARMONICS

Harmonics are often used in the electronic industry. A harmonic of a wave is a component frequency of the signal that is an integer multiple of the fundamental frequency.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

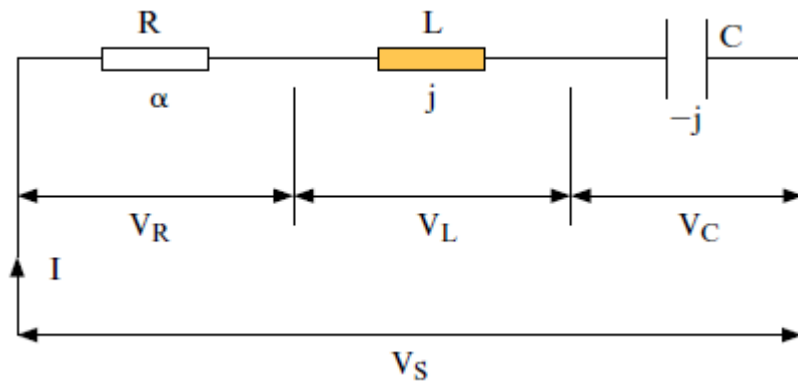
COMPLEX NOTATION

Complex notation is a method used to calculate different quantities in alternating current networks in modulus and angle form, which gives us a much easier method of calculation. Complex notation is very often also referred to as the 'j' notation.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

RLC-NETWORK

Series RLC-networks:



The impedance of the network is mathematically given by:

$$Z = \alpha \pm j\beta \quad (\text{rectangular form})$$

$$= r\sqrt{\theta^\circ} \quad (\text{polar form})$$

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

RESONANCE

Resonance is a condition that will occur in a series or parallel RLC-network when the inductive reactance equals the capacitive reactance.

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

THE Q-FACTORS

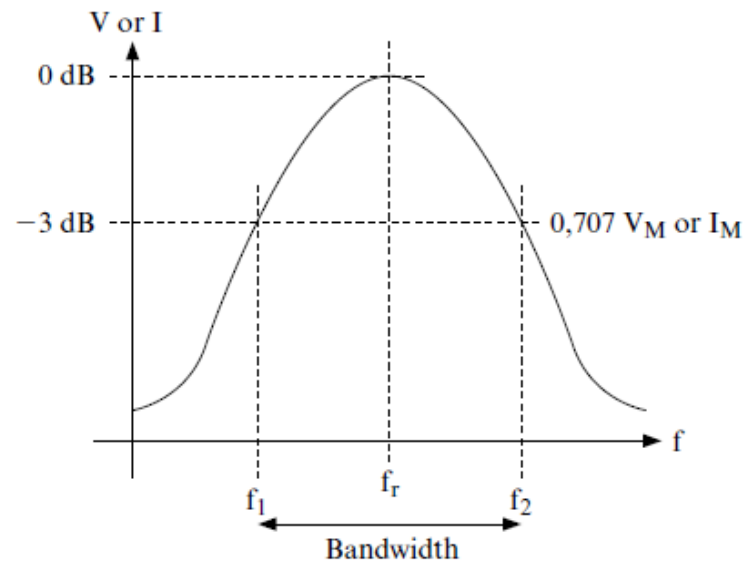
The Q-factor of a network is also termed the 'magnification factor' and is applicable to either a series or parallel resonant network. This factor is mathematically expressed by:

$$Q = \frac{X_L}{R} \text{ or } Q = \frac{1}{R} \left(\frac{L}{C} \right)^{\frac{1}{2}}$$

Module 5: Behaviour of an alternating quantity on RC-, RL- and RLC-circuits (continued)

BANDWIDTH

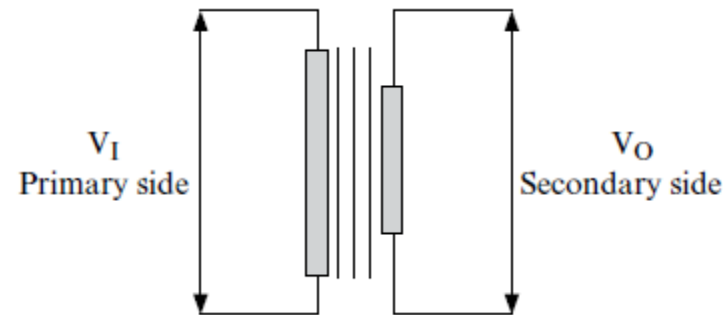
The graphic representation below illustrates the concept of bandwidth:



Module 6: Power supplies

RECTIFIER CONCEPTS

Although different types of transformers are available on the market today, the most common transformer used in conjunction with rectifier networks is the step-down transformer:



Module 6: Power supplies (continued)

FILTER NETWORKS

A filter can be defined as a device or component that will remove the ripple (pulsating) component from the output of a rectifier circuit. There are various types of filters that can be used, such as:

- The capacitor filter;
- The RC-filter; and
- The LC-filter.

Module 6: Power supplies (continued)

NO-LOAD VOLTAGE

The no-load voltage of any power supply may be defined as that voltage which is supplied by the secondary winding of the transformer when the load to that power supply is not connected.

Module 6: Power supplies (continued)

VOLTAGE REGULATION

Voltage regulation may be defined as that change in the output voltage (full-load) for varying load conditions.

Module 6: Power supplies (continued)

VOLTAGE MANIPULATION

Rectifier circuits, full-wave or half-wave, are limited to an output that will never be greater than the peak input voltage to that circuit. It may sometimes be required that greater voltages are required and, for this purpose, we make use of voltage multiplication circuits which may supply two or more times the peak input value as an output.

Module 6: Power supplies (continued)

VOLTAGE REGULATORS

In the field of electronics, it is imperative that the supply voltage supplied to circuits is very stable. For this reason, we must use some form of voltage regulation which is mainly done by one of the following methods:

- Series regulators, or
- Shunt (parallel) regulators.

Module 6: Power supplies (continued)

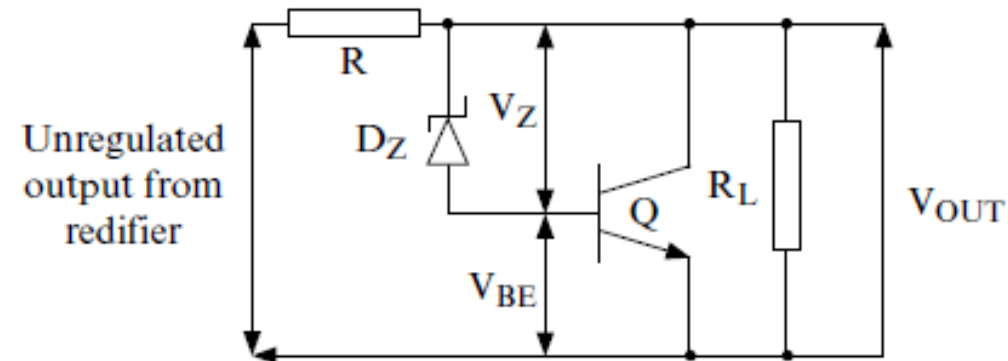
VOLTAGE REFERENCE SOURCES

All Zener diodes are manufactured in order to conduct at a given voltage rating in the reverse-bias area. Zener diodes are mainly used as voltage regulators and are available from 1,5 volts up to 30 volts but higher values can be obtained.

Module 6: Power supplies (continued)

SHUNT (PARALLEL) VOLTAGE REGULATORS

A shunt (parallel) voltage regulator is illustrated below:



Module 6: Power supplies (continued)

ADJUSTABLE POWER SUPPLIES

Adjustable power are mainly used for testing purposes of different circuits within electronic equipment, and there are a number of these regulators available in the market today.

Module 6: Power supplies (continued)

INTEGRATED CIRCUIT RECTIFIERS

An integrated circuit rectifier is a component within which a rectifier is constructed in a single package.

Module 6: Power supplies (continued)

THREE-TERMINAL VOLTAGE REGULATOR

Regulators that are integrated circuit three-terminal voltage regulators are available in a range of voltage and current capabilities. They have a further advantage in that overcurrent and overvoltage protection are already constructed in the integrated circuit package. These regulators are available in the 7800 and 7900 ranges, with a range of voltage and current capabilities.

Module 7: Single-phase control systems

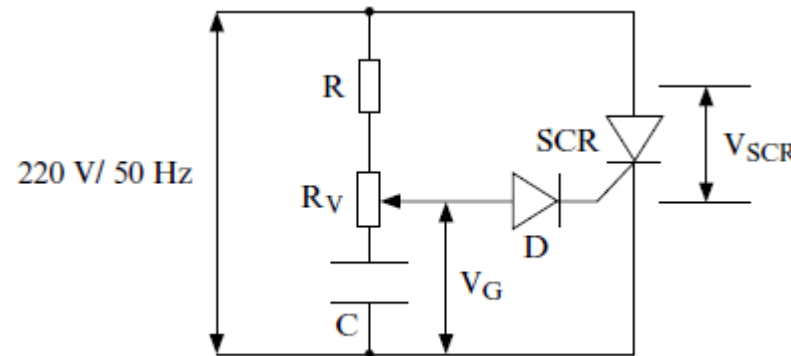
INTRODUCTION

Single-phase control systems are very often also referred to as electronic power control and this, in turn, refers to the control of power to specific electronic devices.

Module 7: Single-phase control systems (continued)

TRIGGER CIRCUIT

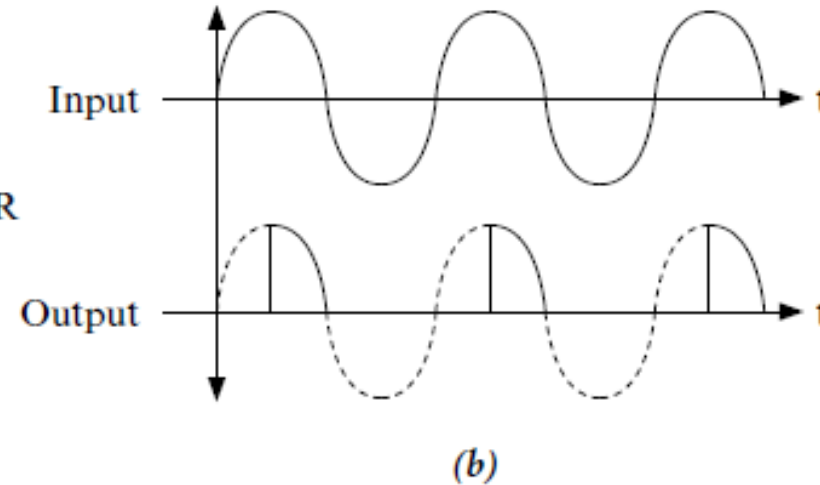
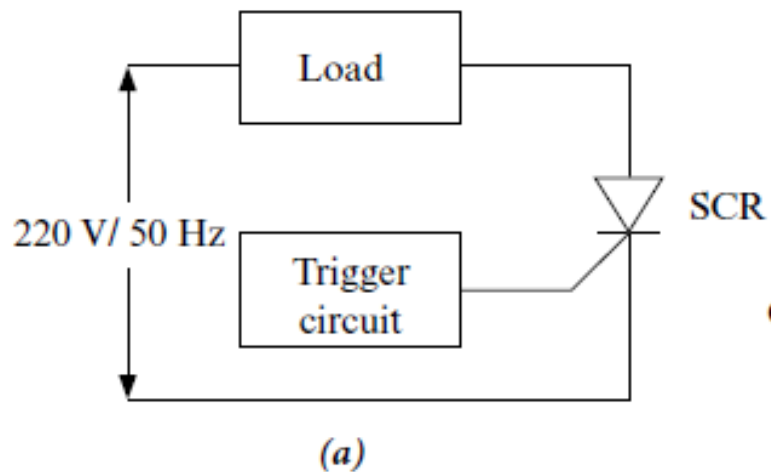
The most important factor when it comes to the design of power control circuits is the design of the trigger circuit. This circuit is responsible to set the SCR, triac or quadrac into conduction. A basic phase control trigger circuit is illustrated below:



Module 7: Single-phase control systems (continued)

SCR CONTROL CIRCUIT (HALF WAVE)

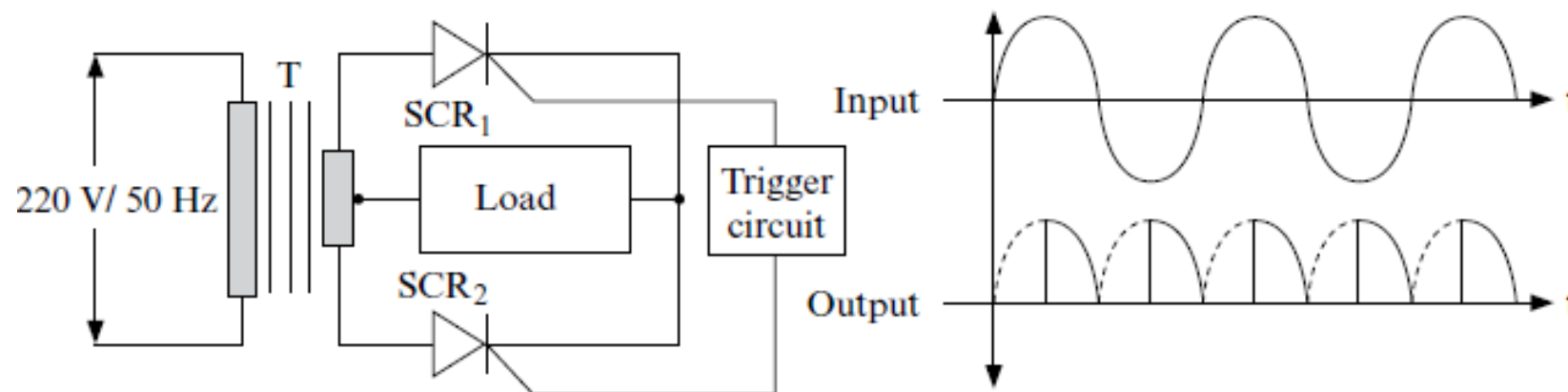
A half-wave dc control circuit, using one SCR and waveforms, is illustrated below:



Module 7: Single-phase control systems (continued)

SCR CONTROL CIRCUIT (FULL WAVE)

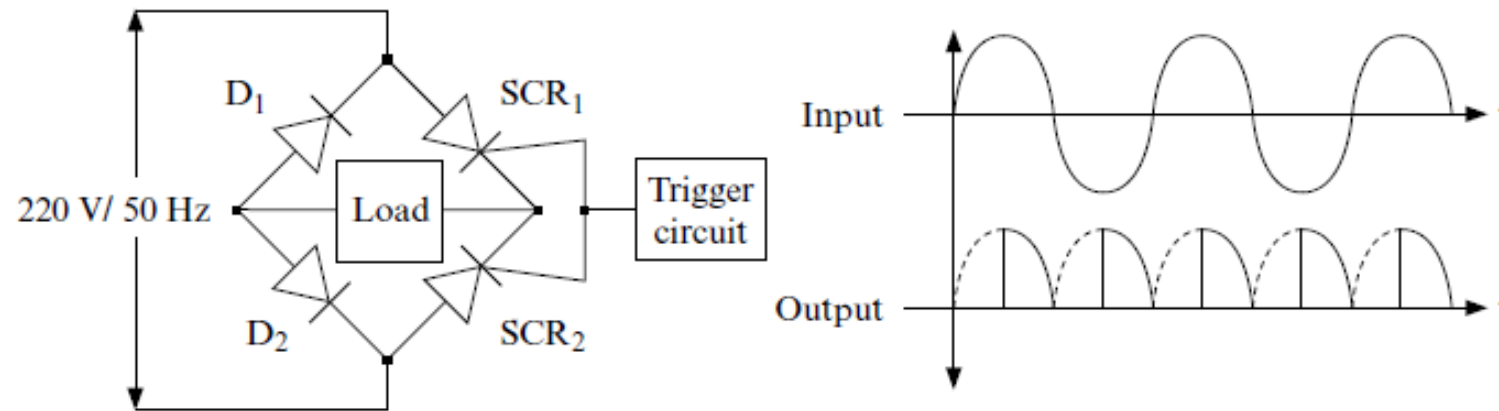
A full-wave dc control circuit, using two SCRs and waveforms, is illustrated below:



Module 7: Single-phase control systems (continued)

SCR CONTROL CIRCUIT (FULL WAVE)

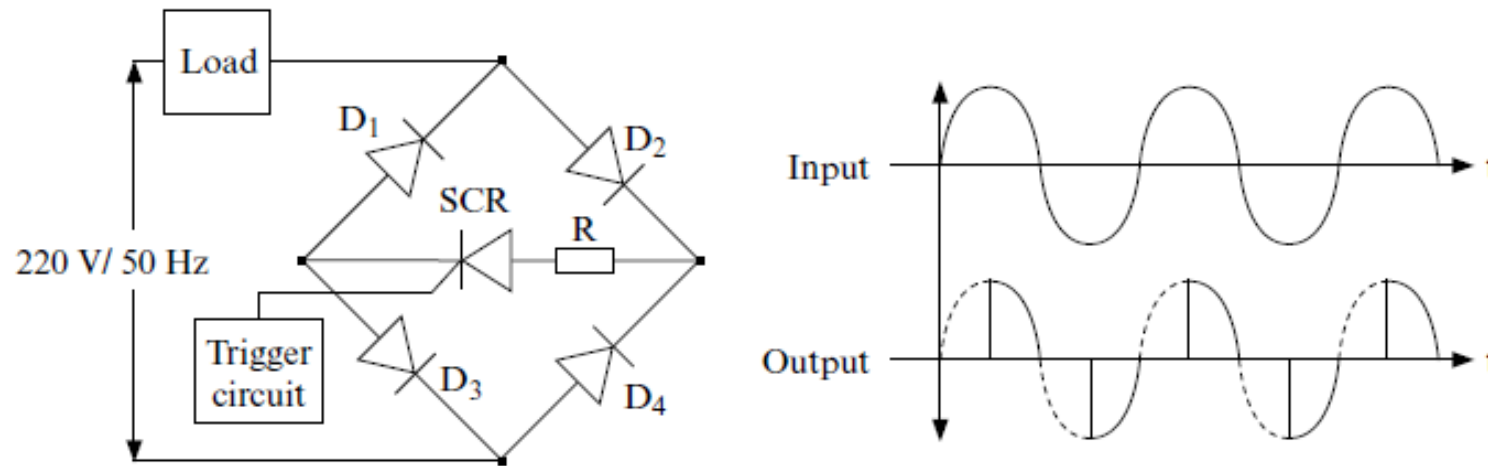
A full-wave dc control circuit, using two SCRs with two diodes in a bridge configuration and waveforms, is illustrated below:



Module 7: Single-phase control systems (continued)

SCR CONTROL CIRCUIT (FULL WAVE AC)

A full-wave ac control circuit, using one SCR with four diodes in a bridge configuration and waveforms, is illustrated below:



Module 7: Single-phase control systems (continued)

OPEN AND CLOSED LOOP SYSTEMS

- An open loop system in power control is a system in which there is no feedback available to adjust the output in a process.
- A closed loop system in power control is a system in which there is feedback available to adjust the output in a process.

Module 8: Transducers

INTRODUCTION

A transducer may be defined as a device that will convert one form of energy into another form of energy.

Module 8: Transducers (continued)

SELECTION OF TRANSDUCERS

The following criteria be taken into account when selecting a transducer.

- Determine the physical quantity that needs to be measured.
- Which transducer principle is best suited to measure a particular quantity?
- What is the level of accuracy that will be required for this measurement?

Module 8: Transducers (continued)

RESISTIVE TRANSDUCERS

A resistive transducer is a device that will exhibit a change in its resistive value when subjected to a change in, inter alia, stress, light, pressure, displacement, etc. All resistive transducers will resort under passive transducers, since they will require an excitation input them i.e. power supplied to them, in order to operate.

Module 8: Transducers (continued)

INDUCTIVE TRANSDUCERS

The operation of inductive transducers is somewhat different from resistive transducers in that the resistance changed with the application of a varying quantity. Inductive transducers make use of a change in magnetic field characteristics and inductive reactance, since it is impossible to alter the inductance value of the inductor.

Module 8: Transducers (continued)

TACHO-GENERATOR

A tacho-generator is a two-phase induction motor and has two coils mounted at right angles to one another. An alternating current is supplied to the one winding and a readout unit, normally a calibrated voltmeter, is connected to the other winding.

Module 8: Transducers (continued)

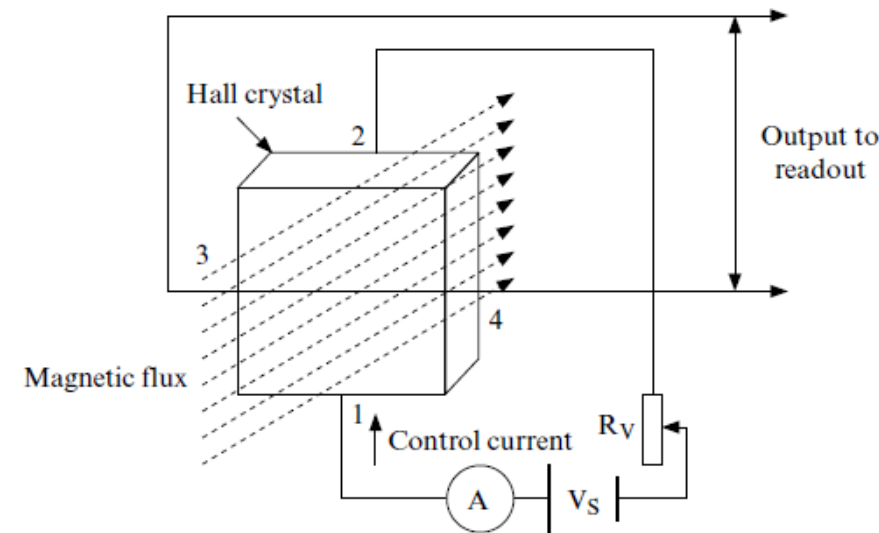
SYNCHRO-SYSTEM

A synchro-system is a system that consists of two parts and they operate as a pair, termed a transmitter and a receiver. This type of system is also termed a remote position control system, since it is possible that the transmitter and receiver are far apart.

Module 8: Transducers (continued)

HALL-EFFECT TRANSDUCERS

The Hall-effect transducer resorts under the heading of magnetic type transducers and is used to measure the strength of a magnetic field. It is illustrated below:



Module 8: Transducers (continued)

PHOTOSENSITIVE TRANSDUCERS

Semiconductors exhibit the phenomena to change their characteristics when exposed to light, whether it be natural light or artificial light, hence the term 'photosensitive', which literally translates to being sensitive to light. These components may operate when they are subjected to a source of light and are then termed photoconductive or photosensitive components, but in this instance they need to be connected in a network or circuit that has a supply voltage.

Module 8: Transducers (continued)

INFRA-RED

Infra-red light is electromagnetic radiation that has a longer wavelength than that of visible light and it cannot be seen but it can be detected. An infra-red detector is simply a transducer of radiant energy, converting radiant energy in the infra-red spectrum into a measurable form.

Module 8: Transducers (continued)

OPTICAL FIBRES

An optical fibre is a flexible, transparent fibre made of high quality extruded glass (silica) or plastic, slightly thicker than a human hair. It can function as a waveguide, or light pipe, to transmit light between the two ends of the fibre. The field of applied science and engineering concerned with the design and application of optical fibres is known as fibre optics.

Module 9: Handling CMOS integrated circuits

PROTECTION

In order to protect the gate-oxide of the CMOS against levels of electrostatic discharge, protective networks can be provided for protection. High-speed CMOS devices, like all MOS devices, have an insulated gate that is subject to voltage breakdown.

Module 9: Handling CMOS integrated circuits (continued)

HANDLING PROCEDURES

In order to safely work with CMOS integrated circuits, it is essential that several precautions should be observed, such as.

- Wrist straps and equipment logs be maintained and audited regularly.
- Do not exceed the maximum ratings specified by the data sheet.
- All low impedance equipment should be connected to CMOS inputs only after the CMOS device is powered.
- Static generating materials shouldn't come in contact with CMOS devices.